SG No. 186 C_{6v}^4 $P6_3mc$ [hexagonal]

* plus set: $+[0,\ 0,\ 0]$

* Wyckoff site: `2a`, site symmetry: `3m.`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `3m.`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ z]$	`[1,2,3,7,8,9]`
2	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[4,5,6,10,11,12]`

Table 2: Wyckoff bond: `6b@2a`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X,\ 2X,\ Z]$	$[0,\ 0,\ z]$	`[1,8]`
2	$[-2X,\ -X,\ Z]$	$[0,\ 0,\ z]$	`[2,7]`
3	$[X,\ -X,\ Z]$	$[0,\ 0,\ z]$	`[3,9]`
4	$[-X,\ -2X,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[4,11]`
5	$[2X,\ X,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[5,10]`
6	$[-X,\ X,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[6,12]`

Table 3: Wyckoff bond: `6c@2a`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$[0,\ 0,\ z]$	`[1,-8]`
2	$[0,\ X,\ 0]$	$[0,\ 0,\ z]$	`[2,-7]`
3	$[-X,\ -X,\ 0]$	$[0,\ 0,\ z]$	`[3,-9]`
4	$[-X,\ 0,\ 0]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[4,-11]`
5	$[0,\ -X,\ 0]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[5,-10]`
6	$[X,\ X,\ 0]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[6,-12]`

Table 4: Wyckoff bond: `12d@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ z]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$[0,\ 0,\ z]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[0,\ 0,\ z]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[6]`
7	$[-Y,\ -X,\ Z]$	$[0,\ 0,\ z]$	`[7]`
8	$[-X+Y,\ Y,\ Z]$	$[0,\ 0,\ z]$	`[8]`
9	$[X,\ X-Y,\ Z]$	$[0,\ 0,\ z]$	`[9]`
10	$[Y,\ X,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[10]`
11	$[X-Y,\ -Y,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[11]`
12	$[-X,\ -X+Y,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[12]`

* Wyckoff site: `2b`, site symmetry: `3m.`

Table 5: Wyckoff bond: `2a@2b`, bond symmetry: `3m.`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3,7,8,9]`
2	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[4,5,6,10,11,12]`

Table 6: Wyckoff bond: `6b@2b`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X,\ 2X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,8]`
2	$[-2X,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[2,7]`
3	$[X,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[3,9]`
4	$[-X,\ -2X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[4,11]`
5	$[2X,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[5,10]`
6	$[-X,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[6,12]`

Table 7: Wyckoff bond: `6c@2b`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,-8]`
2	$[0,\ X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[2,-7]`
3	$[-X,\ -X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[3,-9]`
4	$[-X,\ 0,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[4,-11]`
5	$[0,\ -X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[5,-10]`
6	$[X,\ X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[6,-12]`

Table 8: Wyckoff bond: `12d@2b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[6]`
7	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[7]`
8	$[-X+Y,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[8]`
9	$[X,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[9]`
10	$[Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[10]`
11	$[X-Y,\ -Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[11]`
12	$[-X,\ -X+Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[12]`

* Wyckoff site: `6c`, site symmetry: `.m.`

Table 9: Wyckoff bond: `6a@6c`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, -X, Z]$	$[x, -x, z]$	`[1,7]`
2	$[X, 2X, Z]$	$[x, 2x, z]$	`[2,9]`
3	$[-2X, -X, Z]$	$[-2x, -x, z]$	`[3,8]`
4	$[-X, X, Z]$	$\left[-x, x, z+\frac{1}{2}\right]$	`[4,10]`
5	$[-X, -2X, Z]$	$\left[-x, -2x, z+\frac{1}{2}\right]$	`[5,12]`
6	$[2X, X, Z]$	$\left[2x, x, z+\frac{1}{2}\right]$	`[6,11]`

Table 10: Wyckoff bond: `6b@6c`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, X, 0]$	$[x, -x, z]$	`[1,-7]`
2	$[-X, 0, 0]$	$[x, 2x, z]$	`[2,-9]`
3	$[0, -X, 0]$	$[-2x, -x, z]$	`[3,-8]`
4	$[-X, -X, 0]$	$\left[-x, x, z+\frac{1}{2}\right]$	`[4,-10]`
5	$[X, 0, 0]$	$\left[-x, -2x, z+\frac{1}{2}\right]$	`[5,-12]`
6	$[0, X, 0]$	$\left[2x, x, z+\frac{1}{2}\right]$	`[6,-11]`

Table 11: Wyckoff bond: `12c@6c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, -x, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[x, 2x, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-2x, -x, z]$	`[3]`
4	$[-X, -Y, Z]$	$\left[-x, x, z+\frac{1}{2}\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[-x, -2x, z+\frac{1}{2}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[2x, x, z+\frac{1}{2}\right]$	`[6]`
7	$[-Y, -X, Z]$	$[x, -x, z]$	`[7]`
8	$[-X+Y, Y, Z]$	$[-2x, -x, z]$	`[8]`
9	$[X, X-Y, Z]$	$[x, 2x, z]$	`[9]`
10	$[Y, X, Z]$	$\left[-x, x, z+\frac{1}{2}\right]$	`[10]`
11	$[X-Y, -Y, Z]$	$\left[2x, x, z+\frac{1}{2}\right]$	`[11]`
12	$[-X, -X+Y, Z]$	$\left[-x, -2x, z+\frac{1}{2}\right]$	`[12]`

* Wyckoff site: `12d`, site symmetry: `1`

Table 12: Wyckoff bond: `12a@12d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-Y, X-Y, Z]$	$[-y, x-y, z]$	`[2]`
3	$[-X+Y, -X, Z]$	$[-x+y, -x, z]$	`[3]`
4	$[-X, -Y, Z]$	$\left[-x, -y, z+\frac{1}{2}\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[y, -x+y, z+\frac{1}{2}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[x-y, x, z+\frac{1}{2}\right]$	`[6]`

continued ...

Table 12

No.	vector	center	mapping
7	$[-Y,\, -X,\, Z]$	$[-y,\, -x,\, z]$	[7]
8	$[-X+Y,\, Y,\, Z]$	$[-x+y,\, y,\, z]$	[8]
9	$[X,\, X-Y,\, Z]$	$[x,\, x-y,\, z]$	[9]
10	$[Y,\, X,\, Z]$	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[10]
11	$[X-Y,\, -Y,\, Z]$	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	[11]
12	$[-X,\, -X+Y,\, Z]$	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	[12]